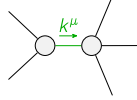
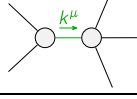
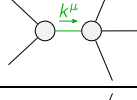
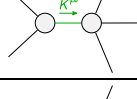
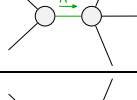
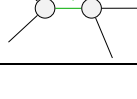
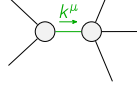
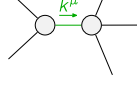


$$\begin{array}{c}
\begin{array}{cc}
\mathcal{K}_{0^+}^{\#1} & \mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{0^+}^{\#1}{}^\dagger & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#1}{}^{\dagger\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & \Upsilon_1 k^2 \\ \hline \end{array}
\end{array}
\begin{array}{cc}
\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#1}{}_{\alpha} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline -\frac{\Upsilon_2 k^2}{36} & \frac{\Upsilon_2 k^2}{18\sqrt{2}} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline \frac{\Upsilon_2 k^2}{18\sqrt{2}} & -\frac{\Upsilon_2 k^2}{18} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cc}
\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{2^+}^{\#1}{}^{\dagger\alpha\beta} & \begin{array}{|c|c|} \hline \frac{3\Upsilon_3 k^2}{4} & 0 \\ \hline \end{array} \\
\mathcal{K}_{2^+}^{\#1}{}^{\dagger\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & \frac{\Upsilon_4 k^2}{2} \\ \hline \end{array}
\end{array}
\end{array}$$

$$\begin{array}{c}
\begin{array}{cc}
\mathcal{J}_{0^+}^{\#1} & \mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{0^+}^{\#1}{}^\dagger & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#1}{}^{\dagger\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & \frac{1}{k^2 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}}} \\ \hline \end{array}
\end{array}
\begin{array}{cc}
\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#1}{}_{\alpha} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline \frac{1}{3 \text{Det}(1^+)} & -\frac{\sqrt{2}}{3 \text{Det}(1^+)} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline -\frac{\sqrt{2}}{3 \text{Det}(1^+)} & \frac{2}{3 \text{Det}(1^+)} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cc}
\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{2^+}^{\#1}{}^{\dagger\alpha\beta} & \begin{array}{|c|c|} \hline \frac{1}{\text{Det}(2^+)} & 0 \\ \hline \end{array} \\
\mathcal{J}_{2^+}^{\#1}{}^{\dagger\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & \frac{1}{\text{Det}(2^+)} \\ \hline \end{array}
\end{array}
\end{array}$$

Abbreviations used in matrices			
$\Upsilon_1 == \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}}$ && $\Upsilon_2 == 6 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}$ && $\Upsilon_3 == 2 \left(\binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_{15}} \right) + \binom{(4)}{\kappa_{16}}$ && $\Upsilon_4 == 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}$ &&			
$\text{Det}(1^+) == -\frac{1}{12} k^2 \left(6 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)$ && $\text{Det}(2^+) == \frac{3}{4} k^2 \left(2 \left(\binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_{15}} \right) + \binom{(4)}{\kappa_{16}} \right)$			
Lagrangian			
$ \begin{aligned} &\binom{(4)}{\kappa_1} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \binom{(4)}{\kappa_{15}} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \frac{1}{2} \binom{(4)}{\kappa_{16}} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta - \\ &\frac{4}{3} \binom{(4)}{\kappa_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta + \frac{5}{9} \binom{(4)}{\kappa_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta - \frac{13}{18} \binom{(4)}{\kappa_{16}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta + \frac{2}{3} \binom{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta + \\ &\frac{2}{9} \binom{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta + \frac{19}{9} \binom{(4)}{\kappa_{16}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - \frac{5}{3} \binom{(4)}{\kappa_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \frac{14}{9} \binom{(4)}{\kappa_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta + \\ &\frac{2}{9} \binom{(4)}{\kappa_{16}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - 2 \binom{(4)}{\kappa_{15}} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta \partial_\delta \mathcal{K}_{\chi\beta}{}^\delta - \binom{(4)}{\kappa_{16}} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta \partial_\delta \mathcal{K}_{\chi\beta}{}^\delta - \frac{1}{3} \binom{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \\ &\frac{10}{9} \binom{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \frac{1}{18} \binom{(4)}{\kappa_{16}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \binom{(4)}{\kappa_{15}} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \binom{(4)}{\kappa_{16}} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned} $			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{J}^\alpha_{\alpha}{}^\beta == 0$	
$\mathcal{J}_{1^+}^{\#2\alpha} == 0$	3	$\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$	
$\mathcal{J}_{1^+}^{\#1\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta_{\beta}{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}_{\beta} == \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$2 \mathcal{J}_{1^+}^{\#1\alpha\beta} + \mathcal{J}_{1^+}^{\#2\alpha\beta} == 0$	3	$\partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\chi \mathcal{J}^{\beta\alpha\chi}$	
Total # constraint(s):	10		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{-6 \binom{(4)}{\kappa_1} - 5 \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)}{\left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)}$
	1	0	$\frac{6 \binom{(4)}{\kappa_1} + 5 \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)}{\left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)}$
	1	0	$- \frac{\sqrt{12 \binom{(4)}{\kappa_1}^2 + 20 \binom{(4)}{\kappa_1} \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) + 11 \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)^2}}{\left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)}$
	1	0	$\frac{\sqrt{12 \binom{(4)}{\kappa_1}^2 + 20 \binom{(4)}{\kappa_1} \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) + 11 \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)^2}}{\left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)}$
	1	0	$- \frac{54 \binom{(4)}{\kappa_1}^2 + 48 \binom{(4)}{\kappa_1} \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) + 11 \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)^2}{\left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(6 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)}$
	1	0	$- \frac{18 \binom{(4)}{\kappa_1}^2 + 24 \binom{(4)}{\kappa_1} \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) + 17 \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)^2}{\left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(6 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)}$
	1	0	$ \begin{aligned} &\left(48 \binom{(4)}{\kappa_1}^2 + 16 \binom{(4)}{\kappa_{15}}^2 + 16 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} + 4 \binom{(4)}{\kappa_{16}}^2 + 32 \binom{(4)}{\kappa_1} \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) - \right. \\ &\left. \sqrt{3} \sqrt{\left(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)^2 \left(396 \binom{(4)}{\kappa_1}^2 + 204 \binom{(4)}{\kappa_1} \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) + 131 \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)^2 \right)} \right) / \\ &\left(\left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(6 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \right) \end{aligned} $
	1	0	$ \begin{aligned} &\left(48 \binom{(4)}{\kappa_1}^2 + 16 \binom{(4)}{\kappa_{15}}^2 + 16 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} + 4 \binom{(4)}{\kappa_{16}}^2 + 32 \binom{(4)}{\kappa_1} \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) + \right. \\ &\left. \sqrt{3} \sqrt{\left(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)^2 \left(396 \binom{(4)}{\kappa_1}^2 + 204 \binom{(4)}{\kappa_1} \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) + 131 \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right)^2 \right)} \right) / \\ &\left(\left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \left(6 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \right) \end{aligned} $
Resolved unitarity condition(s):		(Demonstrably impossible)	